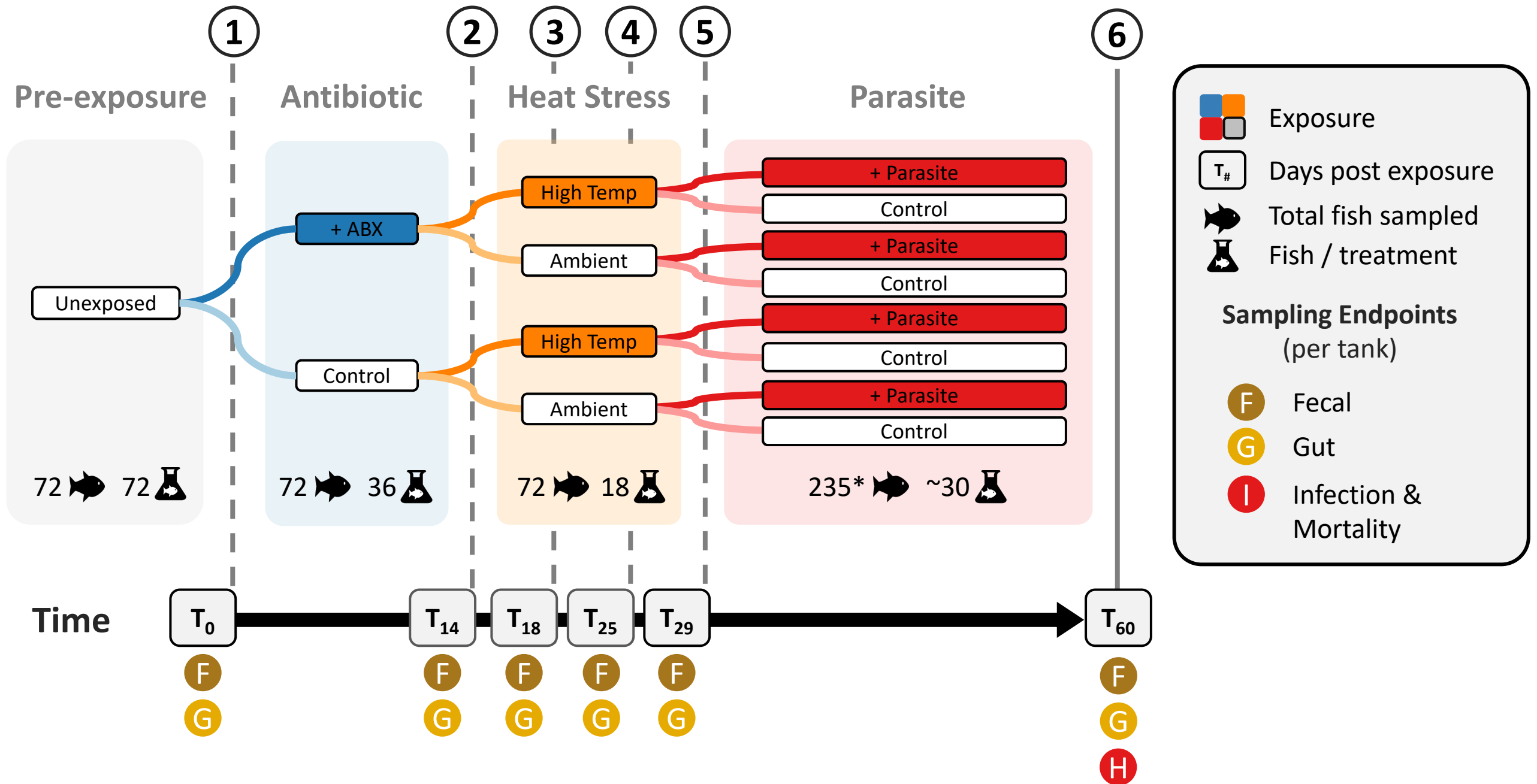


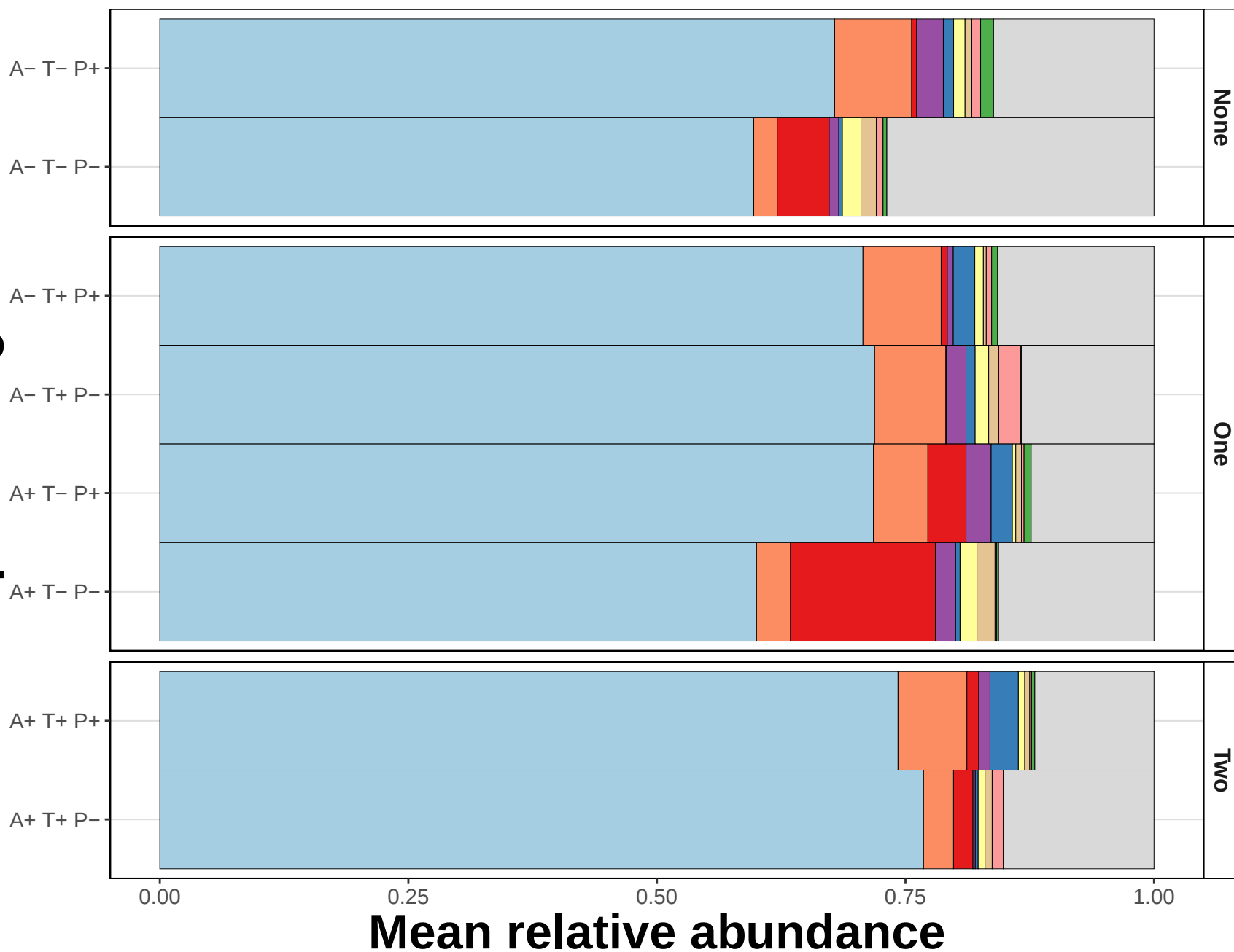


Exposure Regime	Prior Stressor History	Antibiotics	Heat Stress	Parasite
A- T- P-	0	-	-	-
A- T- P+	0	-	-	+
A+ T- P-	1	+	-	-
A+ T- P+	1	+	-	+
A- T+ P-	1	-	+	-
A- T+ P+	1	-	+	+
A+ T+ P-	2	+	+	-
A+ T+ P+	2	+	+	+

Relative Abundance by exposure regime (genus level)

Exposure regime

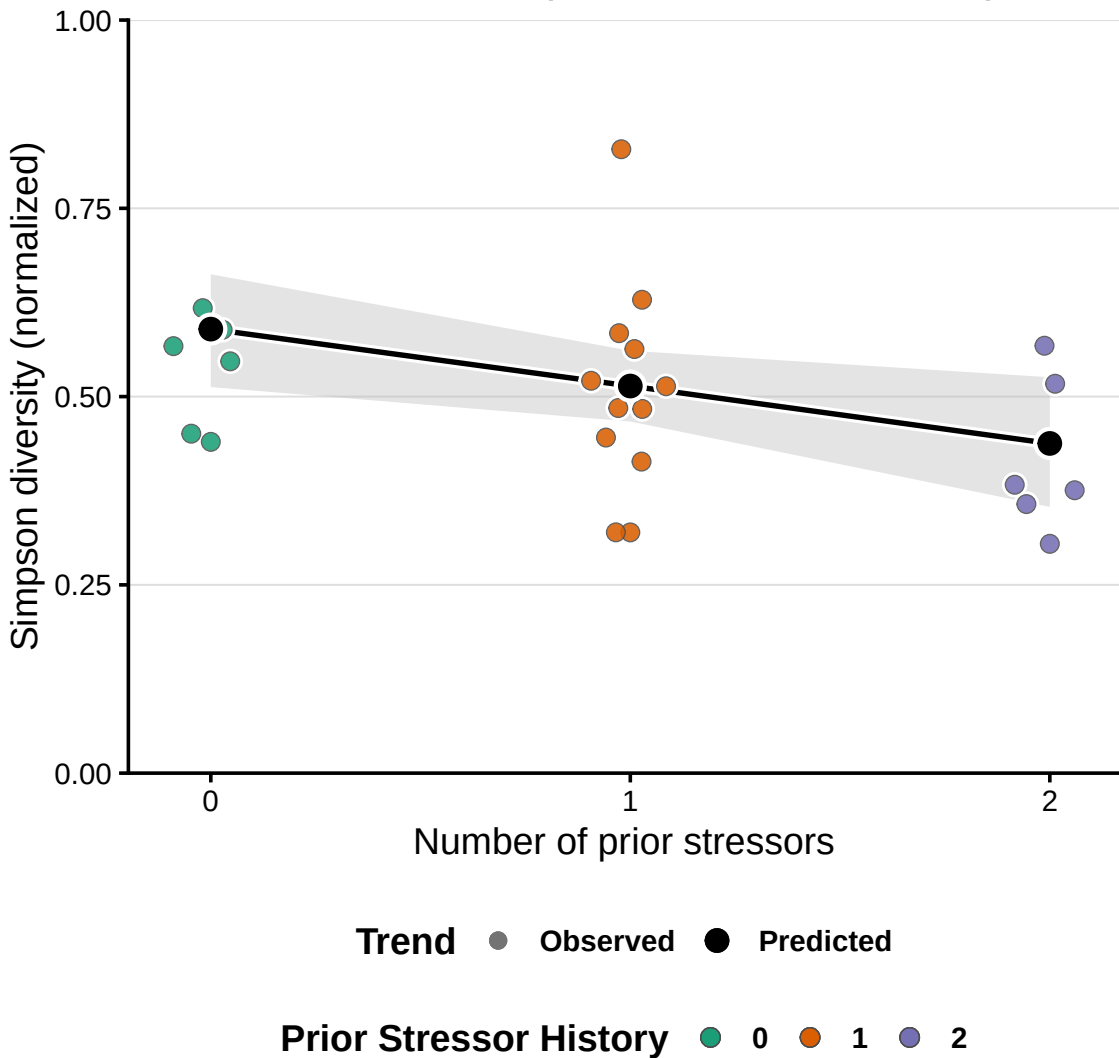
Prior Stressor History



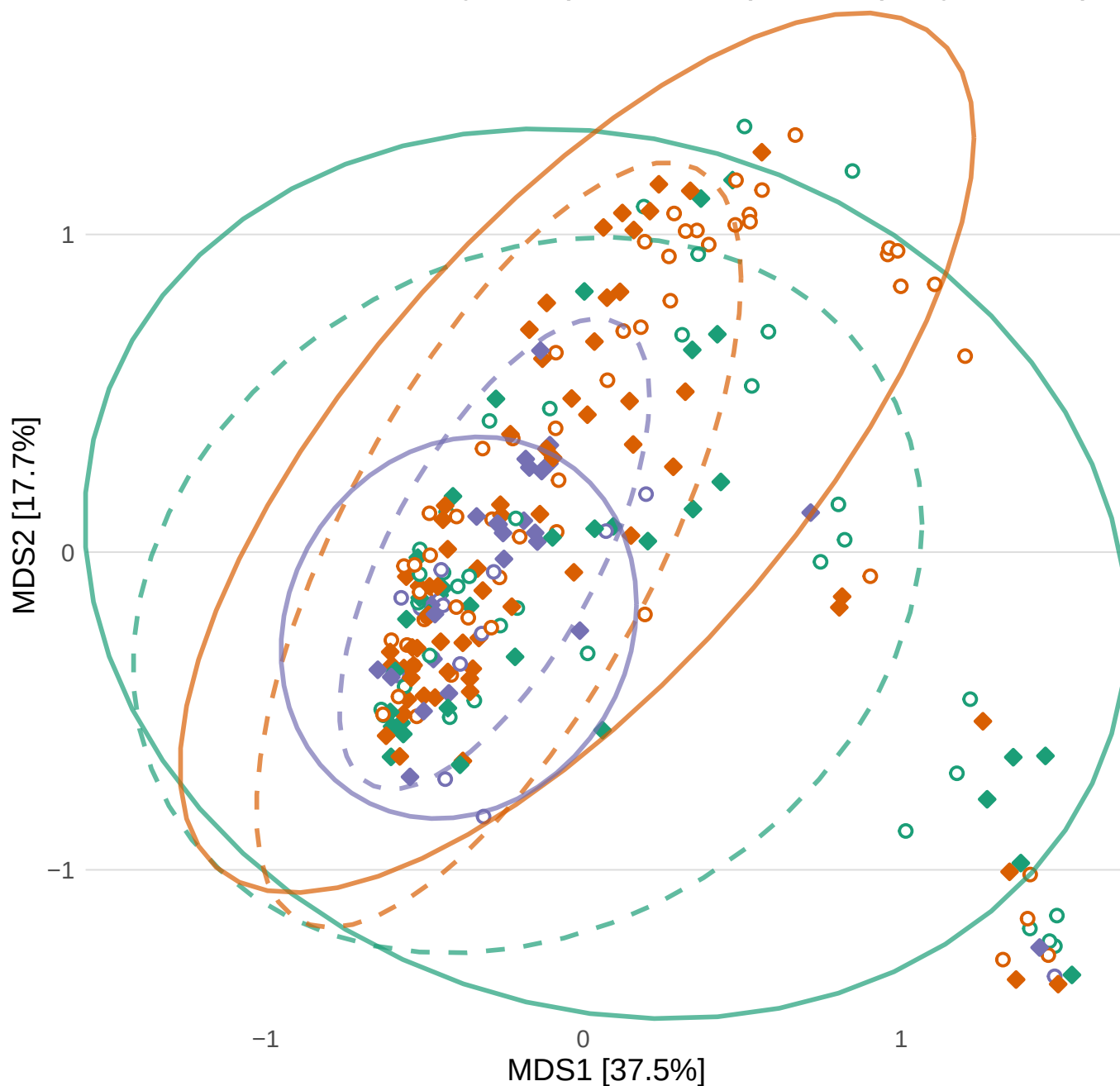
Genus

Aeromonas	Pseudomonas	Culicoidibacter
Cetobacterium	Shewanella	Paracoccaceae Genus
Phreatobacter	Mycobacterium	Flavobacterium
Other		

Cumulative stress exposure effects on Simpson div



PCoA – stressor history and parasite exposure (Bray–Curtis)

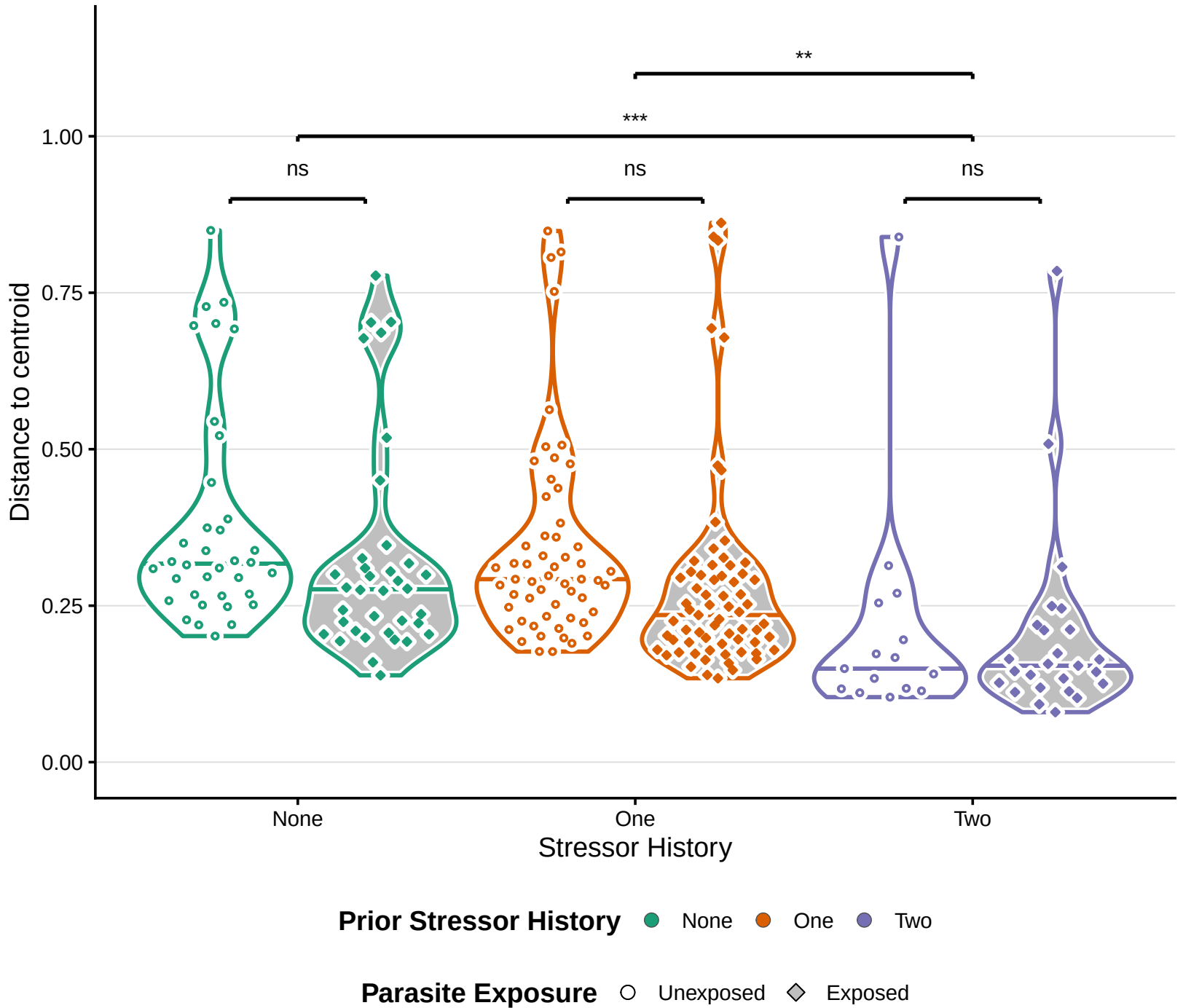


Prior Stressor History ● None ● One ● Two

Parasite Exposure ○ Unexposed ◆ Exposed

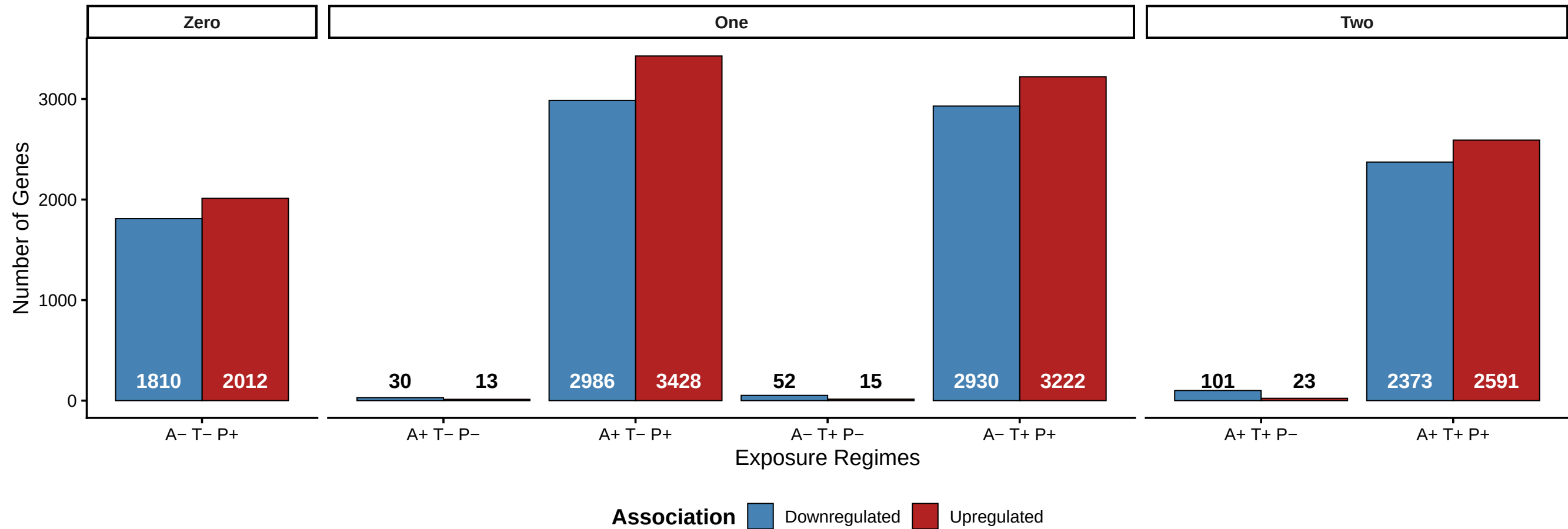
235 samples & 380 taxa (Genus). PCoA tax_transform=identity dist=bray

Beta diversity dispersion by stressor history and parasite exposure (Bray–Curtis)



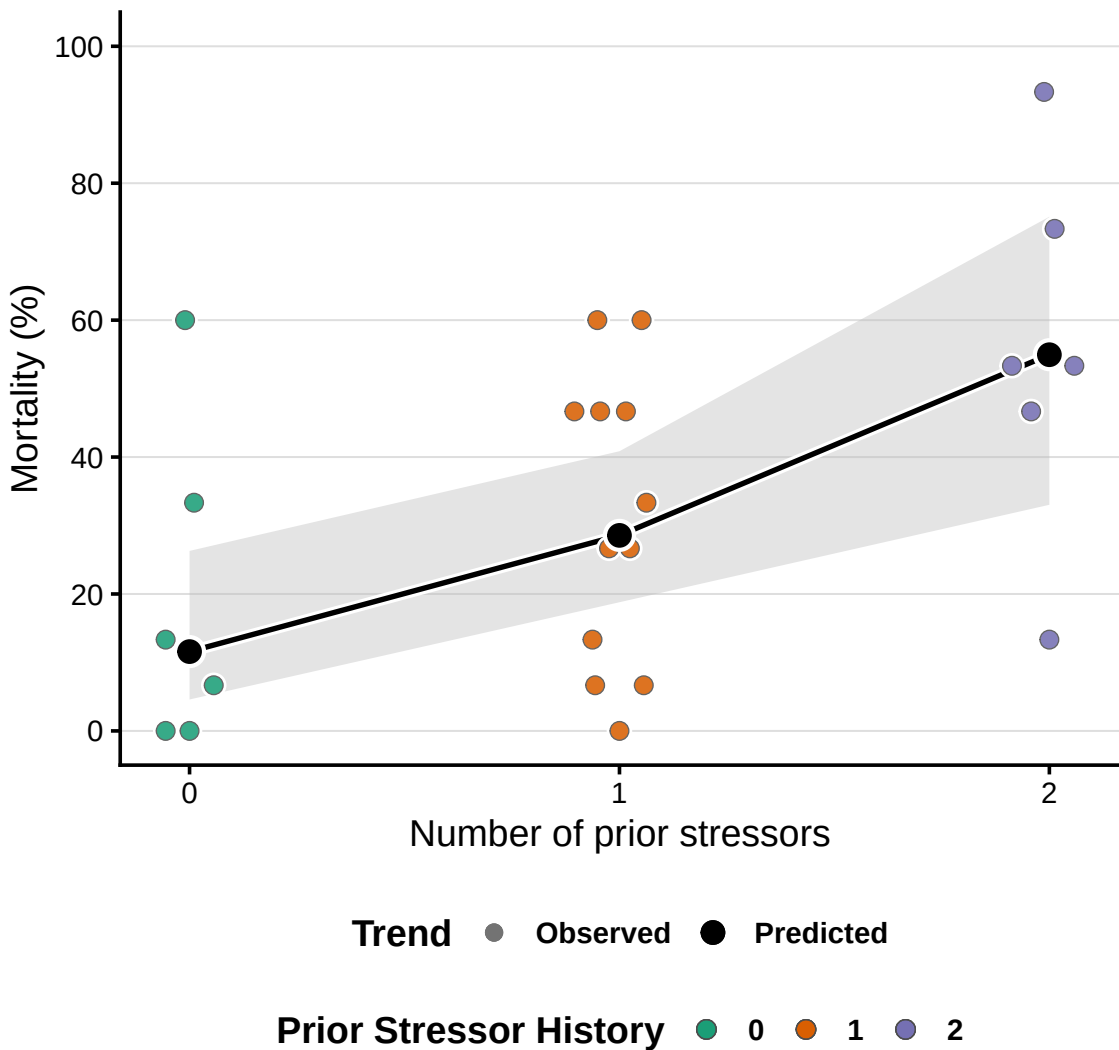
Number of Differentially Expressed Genes by Exposure Regime

Upregulated and downregulated genes ($p_{adj} < 0.05$), ordered by total significant genes within each prior stressor history

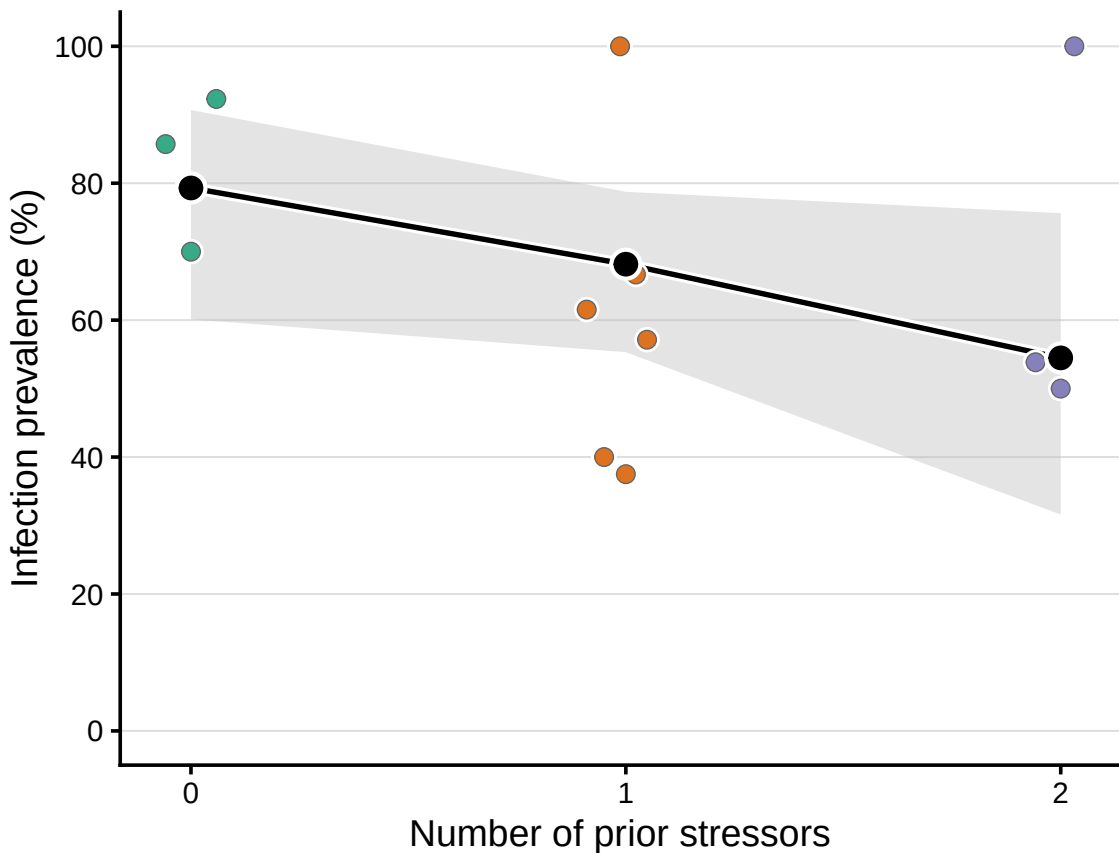


All treatments compared to reference: A- T- P-

Final Mortality by Prior Stressor History



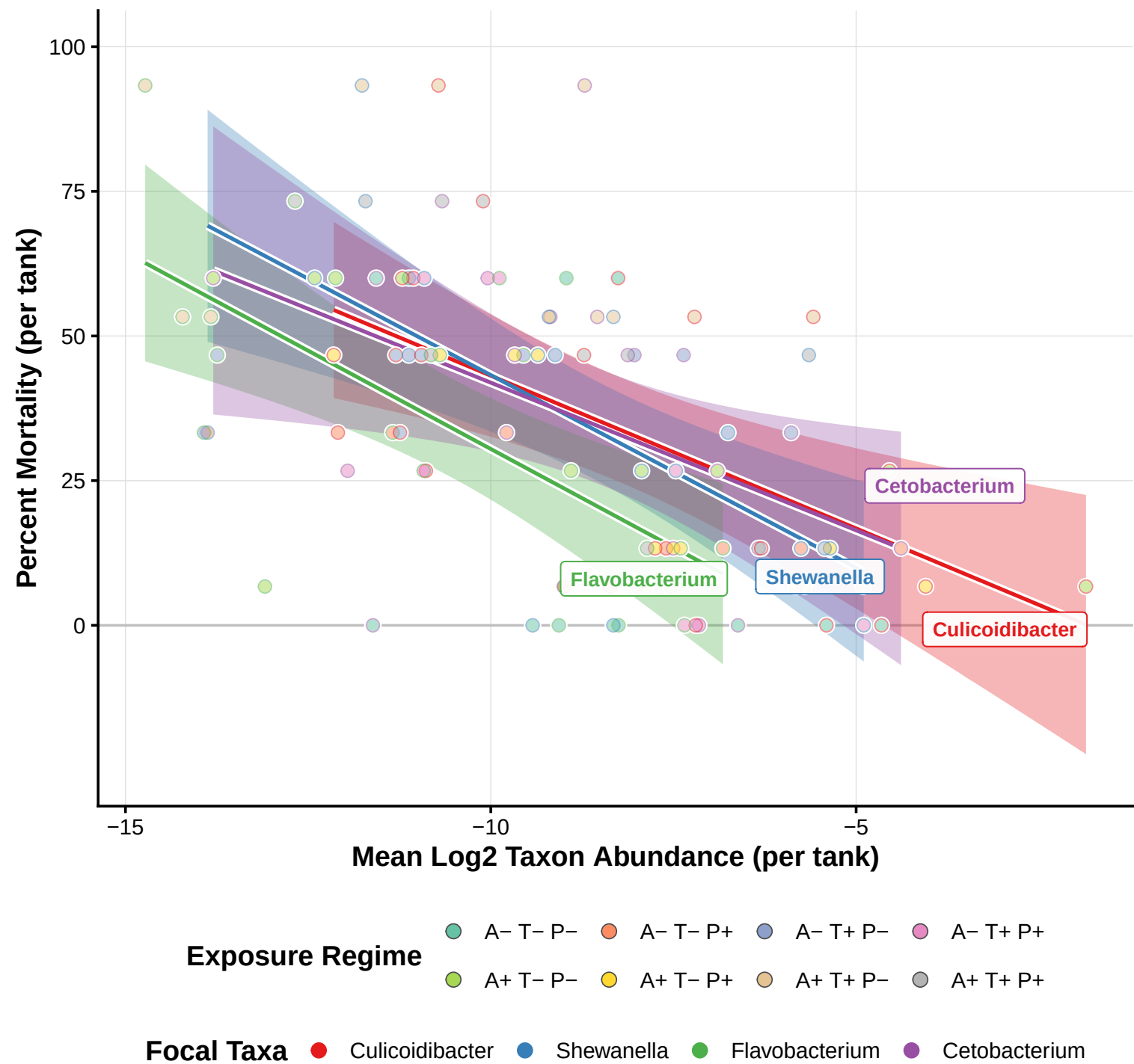
Infection Prevalence by Prior Stressor History



Trend ● **Observed** ● **Predicted**

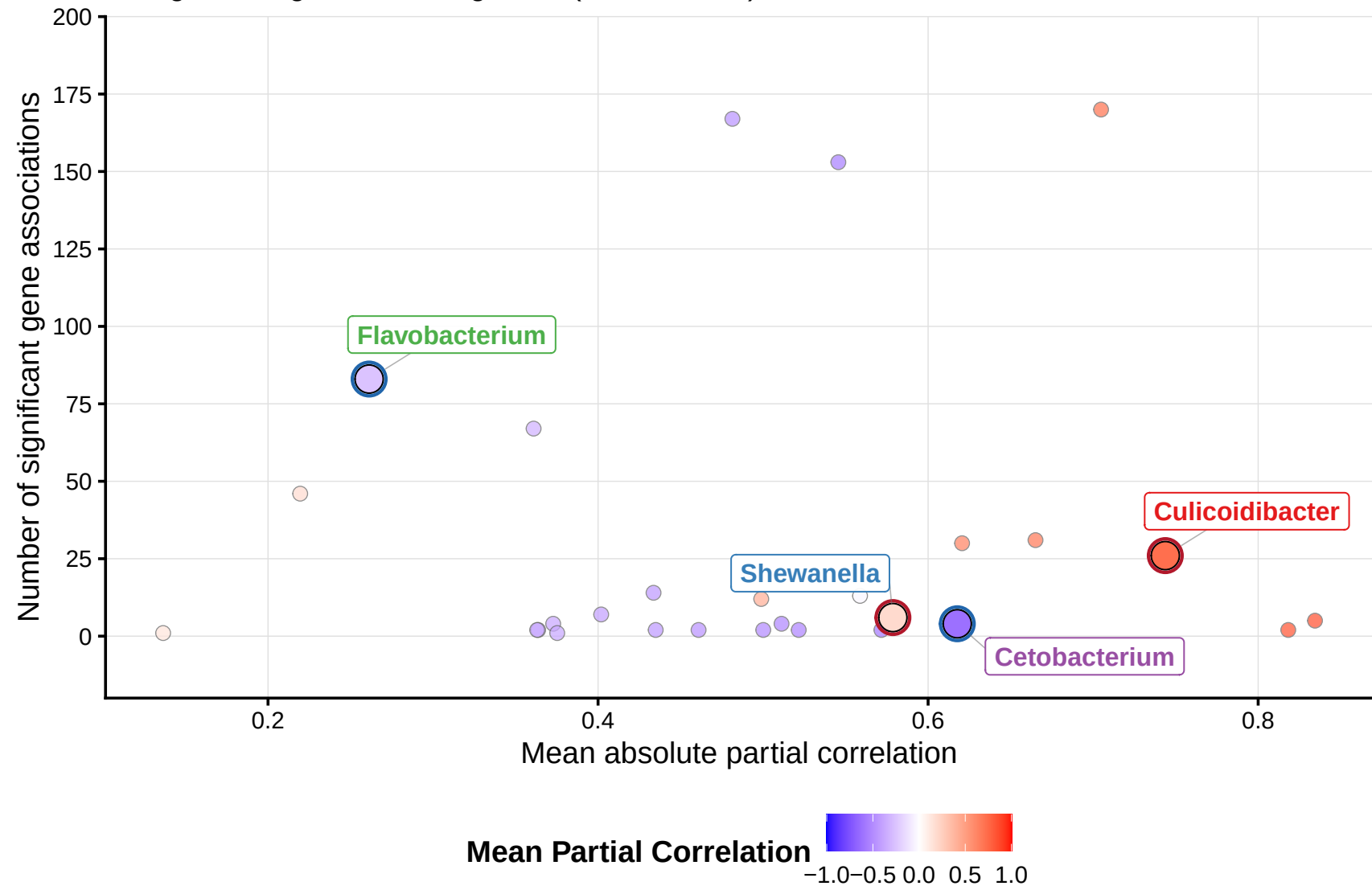
Prior Stressor History ● 0 ● 1 ● 2

Significantly associated genera with host mortality



All taxa with significant partial correlations (FDR < 0.1)

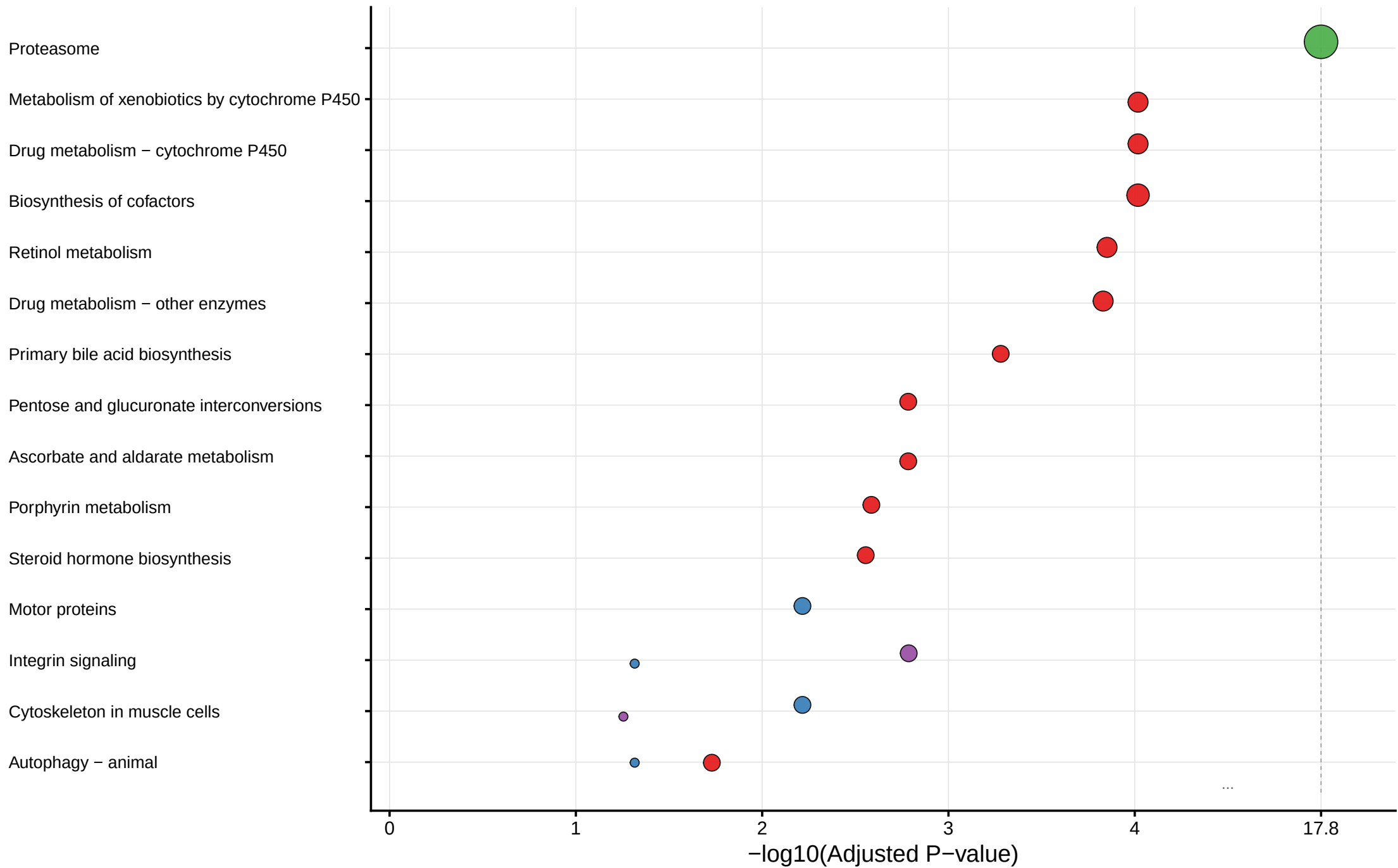
28 significant genera; focal genera (boxed labels)



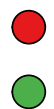
Focal Taxa ● Culicoidibacter ● Shewanella ● Flavobacterium ● Cetobacterium

Functional enrichment – KEGG pathways

Focal genera (combined): genes with significant partial correlations (BH FDR < 0.1); dot fill = focal taxon; x-axis = $-\log_{10}(\text{Adjusted P-value})$



Focal taxa



Culicoidibacter



Shewanella



Flavobacterium



Cetobacterium

Gene count



2



4



6

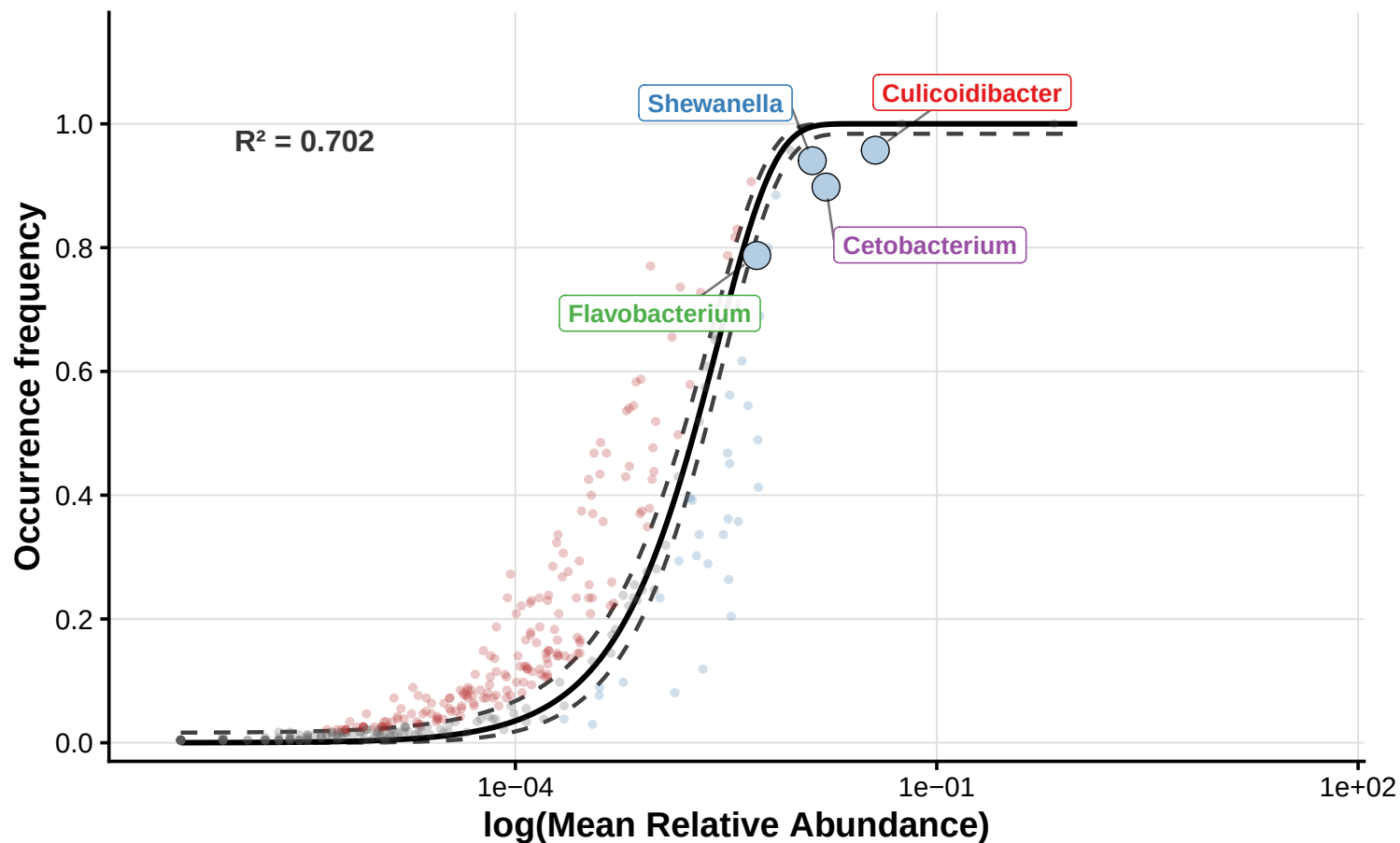


8



10

Sloan Neutral Model (Day 60)



Model Predictions ● Above ● Below ○ Neutral

Focal Taxa ● Culicoidibacter ● Shewanella ● Flavobacterium ● Cetobacterium